I. Linear Algebra

1.1. Def. Vector space over a field, linear ind., span, basis

Def. Dimension.

→ Thm. Dimension is well-defined

Add matrix representation of a vector $x = Ec$

1.2. Def. Subspaces: $W \subset V$ s.t. $\forall x_1, x_2 \in W$, $\forall a \in F$
$x_1 + x_2 \in W$, $ax_1 \in W$

→ Def. Direct sum

2.1 Linear Transformation: $f: X \to Y$, $f \in \mathcal{L}(X, Y)$ also a vectorspace

Def Matrix Representation. choose a basis for X, A, basis B for Y

$f(A) = BM$ or $f(a_1) = B\begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix}$

2.2. Def. Subspaces induced by a linear transformation $f: (X, A) \to (Y, B)$

$\begin{cases} \text{Kernel} & N(f) = \{x \in X : f(x) = 0_Y\} \\ \text{Range} & R(f) = \{y \in Y : f(x) = y\} \end{cases}$ prop: $R(f)$, $N(f)$ are subspaces

2.3. Relation to the matrix

Thm. $f: \underset{m}{X} \to \underset{n}{Y}$, $g: \underset{n}{Y} \to \underset{k}{Z}$ then matrix for $g \circ f$ is $M_g M_f$ gives matrix multiplication

$g(f(A)) = g(BM_f) = CM_gM_f$ $M_f: n \times m$ $M_g: k \times n$

→ Thm. Kernel and Range are related. $f: \underset{m}{(X, A)} \to \underset{n}{(Y, B)}$

Def. Rank(f) = dim$(R(f))$. Then dim$(N(f))$ M_f: $n\begin{bmatrix} & \\ & \end{bmatrix}_m \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$

$\{x : M_f x = 0\} = N(f)$ $R(f) = \{\underset{n \times m}{BM_f} \underset{m \times 1}{c} : c \in F^m\}$

Rank(f) + Nullity$(f) = m$

2.4 Find $N(f)$, $R(f)$ by Gaussian Elimination

Def. ~~permuta~~ Elementary matrices, $PM = LU$ (Reduced Echelon Form)

2.5 Invertible, isomorphism.

3. ~~Inner product~~ Linear Operator

Def. $f \in \mathcal{L}(X, X)$

3.1 Invariant subspace $f(W) \subseteq W$ $\forall x \in W$ $f(x) \in W$

One-dimensional Inv. subspace $f(x) = \lambda x$ $\lambda \in F$

3.2. Matrix: $Mx = \lambda x$ for one-dimensional invariant subspaces.

4. Inner product and duality $(\mathbb{R}^n, \mathbb{C}^n)$

4.1 For $\mathbb{R}^n, \mathbb{C}^n$, $\exists$ ip: $\langle x, y \rangle$: sesqui linear? $\langle x | y \rangle = \overline{\langle y | x \rangle}$

$\bar{z}_1^T z_2$ $z_1^* z_2$ is an inner product

4.2 Cauchy-Schwarz: $\langle x | y \rangle \leq \|x\| \|y\|$

proof. $\|x\|^2 \|y\|^2 = \langle x | x \rangle \langle y | y \rangle$

$\langle x | y \rangle^2$.

4.3. Projection, orthogonal complements, $V = U \oplus U^\perp$

4.4. Parrellolgram law

4.5. Representation theorem $\forall f \in \mathcal{L}(X \to X)$, $\exists x^* \in X$ s.t. $f(x) = \langle x^* | x \rangle$

4.6 Dual? $\langle f(x) | y \rangle = \langle x | f^*(y) \rangle$

5. Spectral Theorems.

Def. Normal matrices

5.2 Applications: power of a matrix, Markov matrix, differential equations

6. Isometry and polar Decomposition, SVD. (positive. unitary)

7. Determinant

7.1 Def. $|A|$ cofactors

property

2.6 Dual Space

4.6 Adjoint, self adjoint operator, normal operator

8. Operator on vector spaces over $\mathbb{C}$ (that not necessarily has an inner product)

9. Operator on vec. space over $\mathbb{R}$

1.1 Theorem: Dimension is well-defined

proof. Let $(e_1, \cdots, e_n)$ $(f_1, \cdots, f_m)$ be two bases of a vector space V.

Because $(e_1, f_1, \cdots, f_m)$ is linearly dependent $e_1 \in \text{span}(f)$

$e_1 = \sum_{j=1}^{m} a_j f_j$ with some nonzero a_{j_1}, then

$\text{span}(e_1, f_{-j_1}) = \text{span}(f)$

Repeat this for all $e_1, \cdots e_n$, if $m < n$, then $\exists e_k$ that

$(e_1, \cdots e_m)$ is a spanning set of vec, but $e_k \notin \text{span}(e_1, \cdots e_m)$ $\forall k > m$

Hence $m \geq n$

Similarly $n \geq m$.

$\square$

1.2 Def. Direct sum $U_1 \oplus \cdots \oplus U_n$ for $U_1, \cdots U_n$ subspaces of V if

$\forall x \in \oplus U_j$, there is only one way to write $x = u_1 + \cdots + u_n$

prop. $U_1 + \cdots + U_m$ is a direct sum iff if $0 = u_1 + \cdots + u_m$, then

$u_1 = u_2 = \cdots = u_m = 0$

$\square$

2.5 Invertible Transformations and Isomorphisms

Def. $f \in \mathcal{L}(X, Y)$ is invertible if $\exists g \in \mathcal{L}(Y, X)$ s.t. $f \circ g = I_Y$ $g \circ f = I_X$

<u>Thm.</u> $f \in \mathcal{L}(X, X)$ is invertible iff $N(f) = 0$ (injective) iff $R(f) = Y$ surjective

<u>Thm.</u> X, Y are isomorphic if $\exists$ invertible $f \in \mathcal{L}(X, Y)$

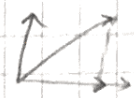
4. Inner Product Space

$\|z\|^2 = z_1\bar{z}_1 + \cdots + z_n\bar{z}_n$ $\qquad \langle x|y\rangle = y^\dagger x$

4.1 Def. $\langle u|v\rangle = \overline{\langle v|u\rangle}$ ~~mean~~ called conjugate symmetry

Def. $u \perp v$ if $\langle u|v\rangle = 0$

<u>Thm.</u> Pythagorean if $\langle u|v\rangle = 0$, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$



<u>Thm.</u> C-S inequality $|\langle u|v\rangle| \le \|u\|\,\|v\|$

proof. $u = \frac{\langle u,v\rangle}{\|v\|^2} v + w$ $\qquad \|u\|^2 = \left(\frac{\langle u,v\rangle}{\|v\|^2}\right)^2 \|v\|^2 + \|w\|^2 \ge \frac{|\langle u,v\rangle|^2}{\|v\|^2}$

<u>Thm.</u> Triangle inequality $\|u+v\| \le \|u\| + \|v\|$

<u>Thm.</u> Parallelogram Equality $\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$

4.2 Gram-Schmidt : Notice it's an upper triangular matrix!

4.3. Every finite-dimensional v.s has an orthonormal basis

4.2. Given $E = [e_1 \ldots e_n]$ $\quad e_1/\|e_1\|$ $\qquad$ GS matrix $\begin{bmatrix} \frac{1}{\|e_1\|} & * & \\ 0 & * & \\ \vdots & & \end{bmatrix}$

$e_2 - \frac{\langle e_1|e_2\rangle}{\|e_1\|^2} e_1$

$\square$

4.4. Riesz Representation $f \in \mathcal{L}(V,F)$. $\exists x^*$ $f(x) = \langle x, x^*\rangle$

proof. If $f \in \mathcal{L}(V,F)$, $f(E) = M$ we can choose E to be o.n.b

then ~~define~~ M is $1\times n$ matrix. $x^* = E M^*$ ← conjugate transpose

then $f(x) = f(EC)$ $\forall x \in V$ $\qquad \langle x, x^*\rangle = ME^*EC = MC$

$= f(E)C = MC =$

4.5 $V = U \oplus U^\perp$ so we have orthogonal proj $\pi: V \to U$ $\qquad$ <u>Oblique proj</u> (IV opposed to OLS)

P_U is ① linear operator ② idempotent projection | ③ ~~symmetric/transition~~ self-adjoint for orthogonality

4.6 Orthogonal proj minimizes distance $\|v - P_U v\| \le \|v - u\|$

5. Spectral Theorems (Linear Operators on Inner Product Spaces)

Def. Adjoint If $T: X \to X$ where X has inner product over $\mathbb{C}$.

Then, $Tx = \langle x | x^* \rangle \ \forall x$ $\quad \langle Tx | y \rangle = \langle x | T^* y \rangle$

Fix y. Tx $\quad TE = EH \quad T^* E = EH^*$

Suppose $T \in \mathcal{L}(V, W)$, T^* $W - V$ s.t. $\langle Tv, w \rangle = \langle v, T^* w \rangle \ \forall v, w$

Thm. $N(T^*)$

$\begin{cases} \text{fundamental subspaces of a linear operator/transformation} \\ \text{vector operations} \\ N(T) = (R(T^*))^\perp \text{ because (1) } N(T) \\ \text{matrix representation if we choose onb} \end{cases}$

Def. Self-Adjoint $T = T^*$

Thm. In $T \in \mathcal{L}(V, \mathbb{C})$, normal T ~~has~~ is diagonalizable, $\exists$ o.n.b consisting of e-vec

$\mathcal{L}(V, \mathbb{R})$ self-adjoint T is diagonalizable.

proof If $TT^* = T^*T$, then consider the null spaces

$N(TT^*) = \{x : TT^* x = 0\} \Rightarrow x \in N(T^*)$ or $T^* x \in N(T)$

For linear operator $R(T) = (N(T))^\perp$ proof

$\forall x \in N(T)$ $\quad Tx = 0 \quad \langle Tx, y \rangle = 0 \ \forall y \in V$

$R(T) \perp N(T)$ $\quad$ If T is self adjoint then $R(T) = (N(T))^\perp$

If $TT^* = T^*T$, for complex vec. space

~~Try~~ $N(T) = R(T^*)^\perp$ $\quad N(TT^*) = R(TT^*)^\perp$ $\quad (TT^*)^* = TT^*$

$N(T) = N(T^*)$

$x \in N(T) \Leftrightarrow$

$\langle Tx, Tx \rangle = 0 \Leftrightarrow \langle x, T^* T x \rangle \Leftrightarrow \langle x, TT^* x \rangle \Leftrightarrow \langle TT^* x, x \rangle = 0$

$\Leftrightarrow \langle T^* x, T^* x \rangle = 0$

~~$N(T - \lambda I) = N(T^* - \bar{\lambda} I)$~~ $\quad$ For normal $N(T) = N(T^*)$

$TT^* = T^* T$

$[(T - \lambda I)(T - \lambda I)^*]^* = \cancel{TT^* - \lambda T - \bar{\lambda} T + \lambda \bar{\lambda} I}$

$= (T - \lambda I)^{**} (T - \lambda I)^*$

$\cancel{(T - \lambda I)^{**} = (T^* - \bar{\lambda} I)^*}$

$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$\cancel{\begin{bmatrix} 1 & \\ & 1 \end{bmatrix} = \lambda^2 \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}}$

Linear Algebra

1. Inner product space

V. $(\cdot,\cdot)$, such that $\langle x|y\rangle = \langle y|x\rangle$, $\langle x+y|z\rangle = \langle x|z\rangle + \langle y|z\rangle$

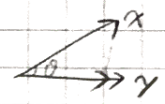
$\langle cx|y\rangle = c\langle x|y\rangle$. □

Thm. If $\langle x|y\rangle = 0$ $(x \perp y)$ then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$

proof $\langle x+y|x+y\rangle = \langle x|y\rangle + \langle x|y\rangle + \overline{\langle y|x\rangle} + \langle y|y\rangle = \|x\|^2 + \|y\|^2$

□

Thm. If $x, y \in V$, $|\langle x|y\rangle| \leq \|x\|\|y\|$



← Cauchy-Schwarz

proof. Consider $x = u + v$ where $u = \frac{y}{\|y\|}\frac{\langle x|y\rangle}{\|y\|}$ $v = x - u$

then $\cancel{\langle u|v\rangle} = \frac{\langle x|y\rangle}{\|y\|^2}\langle y | x - \frac{y}{\|y\|}\frac{\langle x|y\rangle}{\|y\|}\rangle$

$\cancel{\langle v|u\rangle}$

$\langle v|u\rangle = \frac{\langle x|y\rangle}{\|y\|^2}\left[\langle x|y\rangle - \langle x|y\rangle\right] = 0$

Then $\|u+v\|^2 = \|x\|^2 = \|u\|^2 + \|v\|^2$

$= \frac{\langle x|y\rangle^2}{\|y\|^4}\|y\|^2 + \|v\|^2$

$\geq \frac{|\langle x|y\rangle|^2}{\|y\|^2}$

Hence $|\langle x|y\rangle| \leq \|x\|\|y\|$

□

Thm. Parallelogram Law, $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$

proof. $\|x+y\|^2 = \langle x|x\rangle + \langle y|y\rangle + \langle x|y\rangle + \overline{\langle y|x\rangle}$

$\|x-y\|^2 = \langle x|x\rangle + \langle y|y\rangle - \langle x|y\rangle - \overline{\langle y|x\rangle}$

□

Def. <u>Projection</u>. Let $P \in \mathcal{L}(V)$, P: idempotent.

<u>Ort. proj</u> $P \in \mathcal{L}(V)$ idempotent, self-adjoint

← Proj

Orthogonal Proj

Thm 1. (1) If $V = U_1 \oplus U_2$, then $\exists P: V \to U_1$, $v = u_1 + u_2 \mapsto u_1$

it's idempotent

(2) If P is idempotent $\mathcal{L}(V)$, then $\exists U_P$. $V = U_P \oplus U_P'$.

P is the proj of v on U_P.

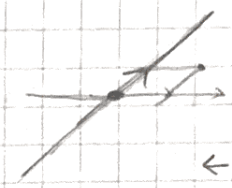
Inner product space

Cauchy Schwarz

Projection

proof. (1) is obvious

(2) Let P be idempotent. Let $U_P = \text{Range}(P)$,



Let $U_P' = N(P)$, we want to show that ← oblique proj

[1] $V = R(P) \oplus N(P)$

[1] $\forall v \in V$, $v = Pv + (v - Pv)$ $Pv \in R(P)$

$P(v - Pv) = 0 \Rightarrow v - Pv \in N(P)$

[2] Then TS $R(P) \cap N(P) = \{0\}$, $x \neq 0$, then if $x \in R(P)$

$x = Py$, $y \neq 0$ $Px = PPy = x \Rightarrow x \notin N(P)$

Hence we have P is a projection (may be oblique)

$\square$

Thm. Orthogonal Projection: idempotent, self-adjoint ← ort. proj

(1) If $V = U \oplus U^{\perp}$, $P: V \to U$ is ort. proj. notice self-adjoint property

(2) if P is ort. proj then $V = R(P) \oplus R^{\perp}(P)$

proof. (1) $P: v \to u + u'$ s.t. $\langle u | u' \rangle = 0$, P is self-adjoint if $\forall x, y$

$\langle Px | y \rangle = \langle x | Py \rangle$ Let $x = Px + (x - Px) = Px + Mx$

$\langle Px | Py + My \rangle = \langle Px | Py \rangle = \langle x | Py \rangle$

(2) $R(P) \perp$ $\forall x \in R(P)$ $\forall y \in N(P)$

$\langle x | y \rangle = \langle Px' | y \rangle = \langle x' | Py \rangle = 0$ hence $x \perp y$

$R(P) \oplus N(P)$

$\square$

R. If a matrix A, B, $\forall x, y \in V = \mathbb{R}^n$

$xAy = xBy$, does it mean $A = B$? Yes, $x \cdot y$ choose basis.

$\square$

Spectral Theorems

1. Adjoints

<u>Thm.</u> Riesz. If $f \in \mathcal{L}(V, F)$, then $\exists x^* \in V$ s.t $f(x) = \langle x | x^* \rangle$ $\forall x \in V$

Def. Let ~~$T \in V$~~ $T \mathcal{L}(X, Y)$, then $\langle Tx | y \rangle$ is a linear functional on X

$\langle Tx | y \rangle = \langle x | T^* y \rangle$

for any given y, define x^* as in Riesz, then $T^* : y \mapsto x^*, y$

is called the adjoint of T.

2. Normal and Self-adjoint operators

Def. $T^*T = TT^*$, $T = T^*$ resp. normal and self-adjoint

If $T^*T = TT^*$ then $\langle Tx | Tx \rangle = \langle T^*x | T^*x \rangle$

TT^* is self adjoint, TT^*TT^* ~~If~~ $\forall x \in N(T^*T)$ $x \in N(TT^*)$

why is adjoint useful $(T - \lambda I)x = 0$ $\langle (T - \lambda I)x | (T - \lambda I)x \rangle = 0$

Let $A = T - \lambda I$, $\langle Ax | Ax \rangle = 0$ find x for an A

If A is self-adjoint then $\langle Ax | Ax \rangle = \langle A^2 x | x \rangle = 0$

If $x \in N(A)$ then $x \in N(A^2)$ ~~so we~~ for $A = A^*$ $\exists$ eigenvalue

$|A| = 0$ $\nexists$ $c_0 A^n x + c_1 + c_0 A^0 = 0$ then λ

For $\langle Ax | Ax \rangle = 0$ $\forall x \in N(A)$ $A = T - \lambda I$

$\langle x | A^* A x \rangle = \langle x | A A^* x \rangle = \overline{\langle A A^* x | x \rangle} = \overline{\langle A^* x | A^* x \rangle}$

$= \langle A^* x | A^* x \rangle = 0$

A_λ $0 = \langle (T - \lambda I)x | (T - \lambda I)x \rangle$

$= \langle Tx | Tx \rangle - \bar{\lambda} \langle Tx | x \rangle - \lambda \overline{\langle Tx | x \rangle} + \lambda \bar{\lambda} \langle x | x \rangle$

$- \bar{\lambda} \langle Tx | x \rangle$ $- \lambda \langle x | Tx \rangle = - \lambda \langle Tx | x \rangle$ $\nearrow x$

$\langle Tx | Tx \rangle - (\lambda + \bar{\lambda}) \langle Tx | x \rangle + \lambda \bar{\lambda} \langle x | x \rangle$ show that

$\exists \lambda$ s.t. $(T - \lambda I)x = 0$ $\cancel{\|Tx\|^2 < (\lambda + \bar{\lambda}) \|Tx\| \|x\|}$

$(\|Tx\| - \lambda \|x\|)(\|Tx\| - \bar{\lambda} \|x\|)$

$- (\lambda + \bar{\lambda}) (\|Tx\| \|x\| - \langle Tx | x \rangle)$